

Maths Refresher

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Machine Learning (CS3306)

January 08, 2026

Basics of Linear Algebra

Vectors and Matrices

- Vectors (column vectors and row vectors) and their transposes

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \quad \mathbf{a}^\top = [a_1 \quad a_2 \quad a_3], \quad \mathbf{b} = [b_1 \quad b_2 \quad b_3], \quad \mathbf{b}^\top = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- Vector with all 0s except a single 1 is called **elementary vector** (or “one-hot” vector in ML)
- Matrix and its transpose (shown for 3×3 matrices)
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$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad \mathbf{A}^\top = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

- For a symmetric matrix (must be square) $\mathbf{A} = \mathbf{A}^\top$
- Diagonal and identity matrices have nonzeros only along the diagonals
- Should know the basic rules of vector addition, matrix addition, etc (won't list here)

Inner Product

- Inner product (or dot product) of two vectors $\mathbf{a} \in \mathbb{R}^D$ and $\mathbf{b} \in \mathbb{R}^D$ is a scalar

$$c = \mathbf{a}^\top \mathbf{b} = \sum_{d=1}^D a_d b_d$$

- Inner product is a measure of similarity of two vectors
- Inner product is zero if $\mathbf{a}$ and $\mathbf{b}$ are orthogonal to each other
- Inner product of two vector of unit length is the same as cosine similarity
- A more general form of inner product: $c = \mathbf{a}^\top \mathbf{M} \mathbf{b}$ (here $\mathbf{M}$ is $D \times D$)
 - $\mathbf{M}$ can be diagonal or full matrix
 - For identity $\mathbf{M}$, it becomes the standard inner product
- Euclidean distance between two vectors can be also written in terms of an inner product

$$d(\mathbf{a}, \mathbf{b}) = \sqrt{(\mathbf{a} - \mathbf{b})^\top (\mathbf{a} - \mathbf{b})} = \sqrt{\mathbf{a}^\top \mathbf{a} + \mathbf{b}^\top \mathbf{b} - 2\mathbf{a}^\top \mathbf{b}}$$

Orthogonal/Orthonormal Vectors and Matrices

- A set of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N$ is called **orthogonal** if

$$\mathbf{a}_i^T \mathbf{a}_j = 0 \quad \forall i \neq j$$

- Moreover, a set of orthogonal vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N$ is called **orthonormal** if

$$\mathbf{a}_i^T \mathbf{a}_i = 1 \quad \forall i$$

- A **matrix** with orthonormal columns is called orthogonal
- For a square orthogonal matrix $\mathbf{A}$, we have $\mathbf{A}\mathbf{A}^T = \mathbf{A}^T\mathbf{A} = \mathbf{I}$

Matrix-Vector/Matrix-Matrix Product as Inner Product

- Important to be conversant with these. Some basic operations worth keeping in mind

- We can 'post-multiply' a matrix by a column vector:

$$Ax = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_1^T x \\ a_2^T x \\ a_3^T x \end{bmatrix}$$

- We can 'pre-multiply' a matrix by a row vector:

$$x^T A = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} x^T a_1 & x^T a_2 & x^T a_3 \end{bmatrix}$$

- In general, we can multiply matrices A and B when the number of columns in A matches the number of rows in B:

$$AB = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_1^T b_1 & a_1^T b_2 & a_1^T b_3 \\ a_2^T b_1 & a_2^T b_2 & a_2^T b_3 \\ a_3^T b_1 & a_3^T b_2 & a_3^T b_3 \end{bmatrix}$$

- We routinely encounter such operations in many ML problems

Outer Product

- **Outer product** of two vectors $\mathbf{a} \in \mathbb{R}^D$ and $\mathbf{b} \in \mathbb{R}^D$ is a matrix. For 3-dim vectors, we'll have

$$\mathbf{C} = \mathbf{a}\mathbf{b}^T = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{bmatrix} \quad (\text{note: } \mathbf{C} \text{ is a rank-1 matrix})$$

- Matrix rank: Linearly indep. number of rows/columns
- Matrix multiplications can also be written as a sum of outer products (sum of rank-1 matrices)

$$\mathbf{A}\mathbf{B}^T = \sum_{k=1}^K \mathbf{a}_k \mathbf{b}_k^T$$

where $\mathbf{a}_k$ and $\mathbf{b}_k$ denote the k -th column of $\mathbf{A}$ (size: $D \times K$) and $\mathbf{B}$ (size: $D \times K$), respectively,

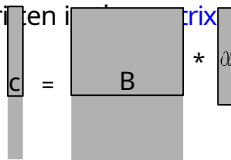
Linear Combination of Vectors as a Matrix-Vector Product

- Linear combination of a set of $D \times 1$ vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_N$ is another vector of the same size

$$\mathbf{c} = \alpha_1 \mathbf{b}_1 + \alpha_2 \mathbf{b}_2 + \dots \alpha_N \mathbf{b}_N$$

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- The α_n 's are scalar-valued combination weights

The above can also be compactly written in **matrix-vector product** form $\mathbf{c} = \mathbf{B}\boldsymbol{\alpha}$


$$\mathbf{c} = \mathbf{B} * \boldsymbol{\alpha}$$

where $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_N]$ is a $D \times N$ matrix, and $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_N]^T$ is an $N \times 1$ column

- vector Note that $\mathbf{c}$ can be also seen as a **linear transformation** of $\boldsymbol{\alpha}$ using $\mathbf{B}$
- Such matrix-vector product are very common in ML problems (especially in **linear models**)

Vector and Matrix Norms

- Roughly speaking, for a vector $\mathbf{x}$, the norm is its “length”
- Some common norms: l_2 norm (Euclidean norm), l_1 norm, l_∞ norm, l_p norm (p

$$\geq \quad \|\mathbf{x}\|_2 = \sqrt{\sum_{n=1}^N x_n^2}, \quad \|\mathbf{x}\|_1 = \sum_{n=1}^N |x_n|, \quad \|\mathbf{x}\|_\infty = \max_{1 \leq n \leq N} |x_n|, \quad \|\mathbf{x}\|_p = \left(\sum_{n=1}^N |x_n|^p \right)^{1/p}$$

- Note: The square of l_2 norm is the inner product of the vector with itself $\sum |x_n|^2$
- Note: $\|\mathbf{x}\|_p$ for $p < 1$ technically not a norm (doesn't satisfy all the formal properties of a norm)

- Nevertheless it is often used in some ML problems (has some interesting properties)
- Norms for a matrix $\mathbf{A}$ (say of size $N \times M$) can also be defined,

e.g.,

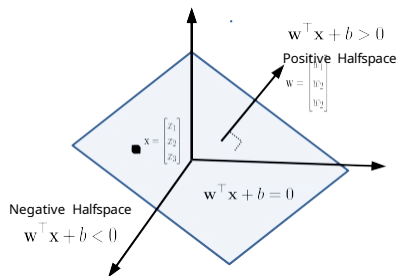
- Frobenius norm: $\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^N \sum_{j=1}^M A_{ij}^2} = \sqrt{\text{trace}(\mathbf{A}^\top \mathbf{A})}$

- Many matrix norms can be written in terms of the singular values of $\mathbf{A}$

Hyperplane

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- An important concept in ML, especially for understanding classification problems
- Divides a vector space into two halves (positive and negative halfspaces)



- Assuming 3-dim space, it can be defined by a vector $w = [w_1, w_2, w_3]$ and
- scalar b w is the vector pointing outward to the hyperplane
- b is the real-valued "bias" if the hyperplane doesn't pass through the origin

Some other things you should know about..

- Eigenvalues, rank, etc. for matrices
- Trace of matrix
- Determinant of matrix (and relation to eigenvalues
- etc) Inverse of matrices
- Positive definite and positive semi-definite matrices (non-negative eigenvalues)
- “Matrix Cookbook” (will provide link) is a nice source of many properties of matrices

Multivariate Calculus and Optimization

Multivariate Calculus and Optimization

- Most of ML problems boil down to solving an optimization problem
- We will usually have to optimize a function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ w.r.t some variable $\mathbf{w} \in \mathbb{R}^D$
- **Gradient** of f w.r.t. $\mathbf{w}$ denotes the direction of steepest change at $\mathbf{w}$, and is defined as

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \vdots \\ \frac{\partial f}{\partial w_D} \end{bmatrix} \quad \text{where} \quad [\nabla f]_i = \frac{\partial f}{\partial w_i}$$

- For multivariate functions $f : \mathbb{R}^D \rightarrow \mathbb{R}^M$, we can likewise define the **Jacobian** matrix

$$\mathbf{J}_f = \begin{bmatrix} \frac{\partial f_1}{\partial w_1} & \cdots & \frac{\partial f_1}{\partial w_D} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_M}{\partial w_1} & \cdots & \frac{\partial f_M}{\partial w_D} \end{bmatrix} \quad \text{where} \quad [\mathbf{J}_f]_{ij} = \frac{\partial f_i}{\partial w_j}$$

- Can also define second derivatives (called **Hessian**): derivative of gradient/Jacobian

Taking Derivatives

- Optimization in ML problems requires being able to take derivatives (i.e., doing Calculus)

What makes it tricky is that usually we are no longer doing optimization w.r.t. a single scalar variable but w.r.t. vectors or sometimes even matrices (thus need [vector/matrix calculus](#))

- For some functions, derivatives are easy (can even be done by hand)

Perhaps the most common, easy ones include derivatives of linear and quadratic functions

$$\nabla_w [X^T w] = X$$
$$\nabla_w [w^T X w] = (X + X^T) w \quad (\text{where } X \text{ is } D \times D$$

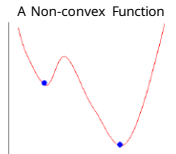
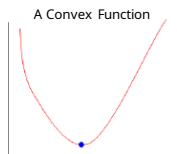
$$\nabla_w [w^T X w] = 2X w \quad (\text{if } X \text{ is symmetric matrix})$$

- The “Matrix Cookbook” contains many derivative formulas (you can use that as a reference even if you don’t know how to compute derivative by hand)
- For more complicated functions, thankfully there exist tool that allow [automatic differentiation](#)

But you should still have a good understanding of derivatives and be familiar with at least some basic results like the above (and some others from the Matrix Cookbook)

Convex Functions

- Convex functions have a unique optima

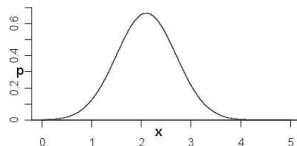
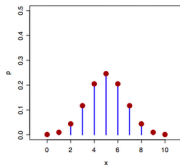


- Optimizing convex functions is usually easier than optimizing non-convex ones
- More on this when we look at optimization for ML later during the semester

Basics of Probability and Probability Distributions

Random Variables

- Informally, a random variable (r.v.) X denotes possible outcomes of an event
- Can be **discrete** (i.e., finite many possible outcomes) or **continuous**



- Some examples of **discrete r.v.**
 - A random variable $X \in \{0, 1\}$ denoting outcomes of a coin-toss
 - A random variable $X \in \{1, 2, \dots, 6\}$ denoting outcome of a dice roll
- Some examples of **continuous r.v.**
 - A random variable $X \in (0, 1)$ denoting the bias of a coin
 - A random variable X denoting heights of students in CS3306
 - A random variable X denoting time to get to your hall from the department

Discrete Random Variables

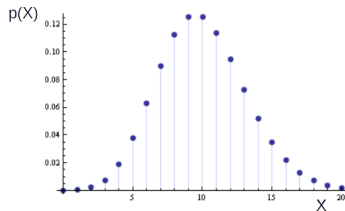
- For a discrete r.v. X , $p(x)$ denotes the probability that $p(X = x)$

$p(x)$ is called the **probability mass function** (PMF)

$$p(x) \geq 0$$

$$p(x) \leq 1$$

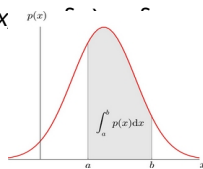
$$\sum_x p(x) = 1$$



Continuous Random Variables

- For a continuous r.v. X , a probability $p(X = x)$ is meaningless
- Instead we use $p(X = x)$ or $p(x)$ to denote the probability density at $X = x$
- For a continuous r.v. X , we can only talk about **probability within an interval** $X \in (x, x + \delta x)$

$p(x)\delta x$ is the probability that $X \in (x, x + \delta x)$



- The probability density $p(x)$ satisfies the following

$$p(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} p(x) dx = 1 \quad (\text{note: for continuous r.v., } p(x) \text{ can be } > 1)$$

A word about notation..

- $p(.)$ can mean different things depending on the context
 - $p(X)$ denotes the distribution (PMF/PDF) of an r.v. X
 - $p(X = x)$ or $p(x)$ denotes the **probability** or **probability density** at point x
 - Actual meaning should be clear from the context (but be careful)
- Exercise the same care when $p(.)$ is a specific distribution (Bernoulli, Beta, Gaussian, etc.) The following means **drawing a random sample** from the distribution $p(X)$

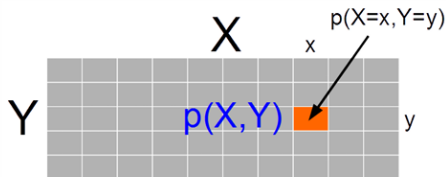
$$x \sim p(X)$$

Joint Probability Distribution

Joint probability distribution $p(X, Y)$ models probability of co-occurrence of two r.v. X, Y

For discrete r.v., the joint PMF $p(X, Y)$ is like a table (that sums to 1)

$$\sum_x \sum_y p(X = x, Y = y) = 1$$

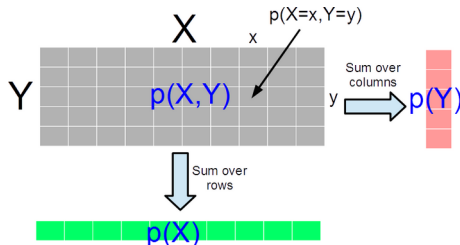


For continuous r.v., we have joint PDF $p(X, Y)$

$$\int_x \int_y p(X = x, Y = y) dx dy = 1$$

Marginal Probability Distribution

- Intuitively, the probability distribution of one r.v. regardless of the value the other r.v. takes
- For discrete r.v.'s: $p(X) = \sum_y p(X, Y = y)$, $p(Y) = \sum_x p(X = x, Y)$
- For discrete r.v. it is the sum of the PMF table along the rows/columns



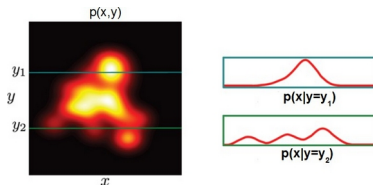
- For continuous r.v.: $p(X) = \int_y p(X, Y = y) dy$, $p(Y) = \int_x p(X = x, Y) dx$
- Note: Marginalization is also called **"integrating out"** (especially in Bayesian learning)

Conditional Probability Distribution

- Probability distribution of one r.v. given the value of the other r.v.
- Conditional probability $p(X | Y = y)$ or $p(Y | X = x)$: like taking a slice of $p(X, Y)$
- For a discrete distribution:



- For a continuous distribution¹:



¹Picture courtesy: Computer vision: models, learning and inference (Simon
Machine Learning (CS3306)

Some Basic

- **Sum rule:** Gives the marginal probability distribution from joint probability distribution
 - For discrete r.v.: $p(X) = \sum_Y p(X, Y)$
 - For continuous r.v.: $p(X) = \int_Y p(X, Y) dY$
- **Product rule:** $p(X, Y) = p(Y|X)p(X) = p(X|Y)p(Y)$
- **Bayes rule:** Gives conditional probability

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$$

- For discrete r.v.: $p(Y|X) = \frac{p(X|Y)p(Y)}{\sum_Y p(X|Y)p(Y)}$
- For continuous r.v.: $p(Y|X) = \frac{p(X|Y)p(Y)}{\int_Y p(X|Y)p(Y) dY}$
- Also remember the **chain rule**

$$p(X_1, X_2, \dots, X_N) = p(X_1)p(X_2|X_1) \dots p(X_N|X_1, \dots, X_{N-1})$$

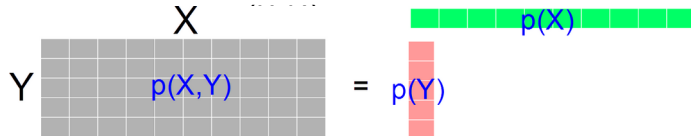
CDF and Quantiles

- Cumulative distribution function (CDF): $F(x) = p(X \leq x)$
- $\alpha \leq 1$ quantile is defined as the x_α s.t.

$$p(X \leq x_\alpha) = \alpha$$

Independence

- X and Y are independent ($X \perp Y$) when knowing one tells nothing about the other

$$\begin{aligned} p(X | Y = y) &= p(X) \\ p(Y | X = x) &= p(Y) \\ p(X, Y) &= p(X)p(Y) \end{aligned}$$


- $X \perp Y$ is also called **marginal independence**
- **Conditional independence** ($X \perp Y | Z$): independence given the value of another r.v. Z

$$p(X, Y | Z = z) = p(X | Z = z)p(Y | Z = z)$$

Expectation

- **Expectation** or **mean** μ of an r.v. with PMF/PDF $p(X)$

$$\mathbb{E}[X] = \sum_x xp(x) \quad (\text{for discrete distributions})$$

$$\mathbb{E}[X] = \int_x xp(x)dx \quad (\text{for continuous distributions})$$

- **Note:** The definition applies to **functions of r.v.** too (e.g., $\mathbb{E}[f(X)]$)
- **Note:** Expectations are always w.r.t. the underlying probability distribution of the random variable involved, so sometimes we'll write this explicitly as $\mathbb{E}_{p_0}[\cdot]$, unless it is clear from the
- context **Linearity of expectation**

$$\mathbb{E}[\alpha f(X) + \beta g(Y)] = \alpha \mathbb{E}[f(X)] + \beta \mathbb{E}[g(Y)]$$

(a very useful property, true even if X and Y are not independent)

- **Rule of iterated/total expectation**

$$\mathbb{E}_{p(X)}[X] = \mathbb{E}_{p(Y)}[\mathbb{E}_{p(X|Y)}[X | Y]]$$

Variance and Covariance

- **Variance** σ^2 (or “spread” around mean μ) of an r.v. with PMF/PDF $p(X)$

$$\text{var}[X] = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2$$

- **Standard deviation:** $\text{std}[X] = \sqrt{\text{var}[X]} = \sigma$
- For two scalar r.v.'s x and y , the **covariance** is defined by

$$\text{cov}[x, y] = \mathbb{E}[\{x - \mathbb{E}[x]\}\{y - \mathbb{E}[y]\}] = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$$

- For **vector** r.v. $\mathbf{x}$ and $\mathbf{y}$, the **covariance matrix** is defined as

$$\text{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[\{\mathbf{x} - \mathbb{E}[\mathbf{x}]\}\{\mathbf{y}^T - \mathbb{E}[\mathbf{y}^T]\}] = \mathbb{E}[\mathbf{x}\mathbf{y}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}^T]$$

- Cov. of components of a vector r.v. $\mathbf{x}$: $\text{cov}[\mathbf{x}] = \text{cov}[\mathbf{x}, \mathbf{x}]$
- **Note:** The definitions apply to functions of r.v. too (e.g., $\text{var}[f(X)]$)
- **Note:** Variance of sum of independent r.v.'s: $\text{var}[X + Y] = \text{var}[X] + \text{var}[Y]$

KL Divergence

- KullbackLeibler divergence between two probability distributions $p(X)$ and $q(X)$

$$KL(p||q) = \int p(X) \log \frac{p(X)}{q(X)} dX = - \int p(X) \log \frac{q(X)}{p(X)} dX \quad (\text{for continuous distributions})$$

$$KL(p||q) = \sum_{k=1}^K p(X = k) \log \frac{p(X = k)}{q(X = k)} \quad (\text{for discrete distributions})$$

- It is non-negative, i.e., $KL(p||q) \geq 0$, and zero if and only if $p(X)$ and $q(X)$ are the same
- For some distributions, e.g., Gaussians, KL divergence has a closed form expression
- KL divergence is not symmetric, i.e., $KL(p||q) \neq KL(q||p)$

Entropy

- Entropy of a continuous/discrete distribution $p(X)$

$$H(p) = - \int p(X) \log p(X) dX$$

$$H(p) = - \sum_{k=1}^K p(X = k) \log p(X = k)$$

- In general, a peaky distribution would have a smaller entropy than a flat distribution
- Note that the KL divergence can be written in terms of expectation and entropy terms

$$KL(p||q) = \mathbb{E}_{p(X)}[-\log q(X)] - H(p)$$

- Some other definition to keep in mind: conditional entropy, joint entropy, mutual information, etc.

Transformation of Random Variables

Suppose $\mathbf{y} = f(\mathbf{x}) = \mathbf{Ax} + \mathbf{b}$ be a linear function of an r.v. $\mathbf{x}$

Suppose $E[\mathbf{x}] = \boldsymbol{\mu}$ and $\text{cov}[\mathbf{x}] = \boldsymbol{\Sigma}$

- Expectation of $\mathbf{y}$

$$E[\mathbf{y}] = E[\mathbf{Ax} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$$

- Covariance of $\mathbf{y}$

$$\text{cov}[\mathbf{y}] = \text{cov}[\mathbf{Ax} + \mathbf{b}] = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$$

Likewise if $y = f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} + b$ is a scalar-valued linear function of an r.v. $\mathbf{x}$:

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- $E[y] = E[\mathbf{a}^T \mathbf{x} + b] = \mathbf{a}^T \boldsymbol{\mu} + b$

$$\text{var}[y] = \text{var}[\mathbf{a}^T \mathbf{x} + b] = \mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a}$$

Another very useful property
worth remembering

Common Probability Distributions

Important: We will use these extensively to model **data** as well as **parameters**

Some **discrete distributions** and what they can model:

- **Bernoulli:** Binary numbers, e.g., outcome (head/tail, 0/1) of a coin toss
- **Binomial:** Bounded non-negative integers, e.g., # of heads in n coin tosses
- **Multinomial:** One of K (>2) possibilities, e.g., outcome of a dice roll
- **Poisson:** Non-negative integers, e.g., # of words in a document
- .. and many others

Some **continuous distributions** and what they can model:

- **Uniform:** numbers defined over a fixed range
- **Beta:** numbers between 0 and 1, e.g., probability of head for a biased coin
- **Gamma:** Positive unbounded real numbers
- **Dirichlet:** vectors that sum of 1 (fraction of data points in different clusters)

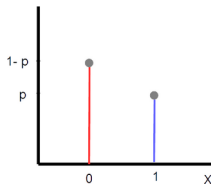
Discrete Distributions

Bernoulli Distribution

- Distribution over a binary r.v. $x \in \{0, 1\}$, like a coin-toss outcome
- Defined by a probability parameter $p \in (0, 1)$

$$P(x = 1) = p$$

- Distribution defined as: $\text{Bernoulli}(x ; p) = p^x (1 - p)^{1-x}$



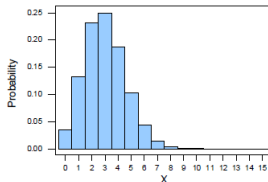
- Mean: $E[x] = p$
- Variance: $\text{var}[x] = p(1 - p)$

Binomial Distribution

- Distribution over number of successes m (an r.v.) in a number of trials
- Defined by two parameters: total number of trials (N) and probability of each success $p \in (0, 1)$
- Can think of Binomial as multiple independent Bernoulli trials
- Distribution defined as

$$\text{Binomial}(m; N, p) = \binom{N}{m} p^m (1 - p)^{N-m}$$

Binomial distribution with $n = 15$ and $p = 0.2$



- Mean: $E[m] = Np$
- Variance: $\text{var}[m] = Np(1 - p)$

Multinoulli Distribution

- Also known as the **categorical distribution** (models categorical variables)
- Think of a random assignment of an item to one of K bins - a K dim. binary r.v. $\mathbf{x}$ with single $\sum_{k=1}^K x_k = 1$): **Modeled by a multinoulli**

$$\underbrace{[0 \ 0 \ 0 \ \dots 0 \ 1 \ 0 \ 0]}_{\text{length} = K}$$

- Let vector $\mathbf{p} = [p_1, p_2, \dots, p_K]$ define the probability of going to each bin
 - $p_k \in (0, 1)$ is the probability that $x_k = 1$ (assigned to bin k)
 - $\sum_{k=1}^K p_k = 1$
- The multinoulli is defined as: $\text{Multinoulli}(\mathbf{x}; \mathbf{p}) = \prod_{k=1}^K p_k^{x_k}$
- Mean: $\mathbb{E}[x_k] = p_k$
- Variance: $\text{var}[x_k] = p_k(1 - p_k)$

Multinomial Distribution

- Think of repeating the Multinoulli N times

- Like distributing N items to K bins. Suppose x_k is count in bin k

$$0 \leq x_k \leq N$$

$$\sum_{k=1}^K x_k = N$$

- $\forall k = 1, \dots, K,$

- Multinomial models the bin allocations via a discrete vector $\mathbf{x}$ of size K . Assume probability of going to each bin: $\mathbf{p} = [p_1, p_2, \dots, p_K]$

- Distribution defined as

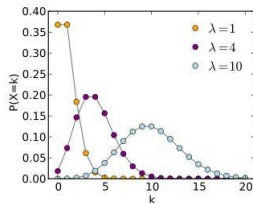
$$\text{Multinomial}(\mathbf{x}; N, \mathbf{p}) = \binom{N}{x_1 x_2 \dots x_K} \prod_{k=1}^K p_k^{x_k}$$

- Mean: $E[x_k] = Np_k$
- Variance: $\text{var}[x_k] = Np_k(1 - p_k)$
- Note: For $N = 1$, multinomial is the same as multinoulli

Poisson Distribution

- Used to model a non-negative integer (count) r.v. k
- Examples: number of words in a document, number of events in a fixed interval of time, etc. Defined by a positive rate parameter λ
- Distribution defined as

$$\text{Poisson}(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \quad k = 0, 1, 2, \dots$$



- Mean: $E[k] = \lambda$
- Variance: $\text{var}[k] = \lambda$

The Empirical Distribution

- Given a set of points $\varphi_1, \dots, \varphi_K$, the empirical distribution is a discrete distribution defined as

$$p_{emp}(A) = \frac{1}{K} \sum_{k=1}^K \delta_{\varphi_k}(A)$$

where $\delta_{\varphi}(\cdot)$ is the **dirac function** located at φ , s.t.

$$\delta_{\varphi}(A) = \begin{cases} 1 & \text{if } \varphi \in A \\ 0 & \text{if } \varphi \notin A \end{cases}$$

- The “weighted” version of the empirical distribution is

$$p_{emp}(A) = \sum_{k=1}^K w_k \delta_{\varphi_k}(A) \quad \left(\text{where } \sum_{k=1}^K w_k = 1 \right)$$

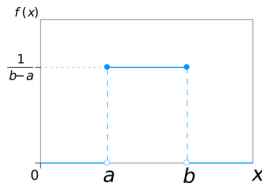
and the weights and points (w_k, φ_k , $k=1, \dots, K$) together define this discrete distribution

Continuous Distributions

Uniform Distribution

- Models a continuous r.v. x distributed uniformly over a finite interval $[a, b]$

$$\text{Uniform}(x; a, b) = \frac{1}{b - a}$$

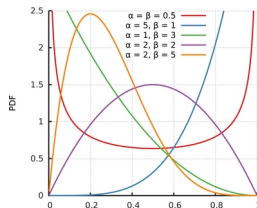


- Mean: $E[x] = \frac{(b+a)}{2}$
- Variance: $\text{var}[x] = \frac{(b-a)^2}{12}$

Beta Distribution

- Used to model an r.v. p between 0 and 1 (e.g., a probability)
- Defined by two **shape parameters** α and β

$$\text{Beta}(p; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1} (1 - p)^{\beta-1}$$

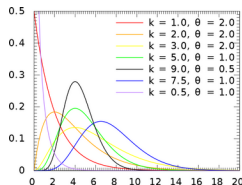


- Mean: $E[p] = \frac{\alpha}{\alpha + \beta}$
- Variance: $\text{var}[p] = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$
- Often used to model the probability parameter of a Bernoulli or Binomial (also **conjugate** to these distributions)

Gamma Distribution

- Used to model positive real-valued r.v. x
- Defined by a **shape parameters** k and a **scale parameter** θ

$$\text{Gamma}(x; k, \theta) = \frac{x^{k-1} e^{-\frac{x}{\theta}}}{\theta^k \Gamma(k)}$$



- Mean: $E[x] = k\theta$
- Variance: $\text{var}[x] = k\theta^2$
- Often used to model the rate parameter of Poisson or exponential distribution (conjugate to both), or to model the inverse variance (precision) of a Gaussian (conjugate to Gaussian if mean known)

Note: There is another equivalent parameterization of gamma in terms of **shape** and **rate** parameters (rate = 1/scale). Another related distribution: Inverse gamma

Dirichlet Distribution

- Used to model non-negative r.v. vectors $\mathbf{p} = [p_1, \dots, p_K]$ that sum to 1

$$0 \leq p_k \leq 1, \quad \forall k = 1, \dots, K \quad \text{and} \quad \sum_{k=1}^K p_k = 1$$

- Equivalent to a distribution over the $K - 1$ dimensional simplex
- Defined by a K size vector $\alpha = [\alpha_1, \dots, \alpha_K]$ of positive reals

- Distribution defined as

$$\text{Dirichlet}(\mathbf{p}; \alpha) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K p_k^{\alpha_k - 1}$$

- Often used to model the probability vector parameters of Multinoulli/Multinomial distribution
- Dirichlet is conjugate to Multinoulli/Multinomial
- Note:** Dirichlet can be seen as a generalization of the Beta distribution. Normalizing a bunch of Gamma r.v.'s gives an r.v. that is Dirichlet distributed.

Dirichlet Distribution

- For $\mathbf{p} = [p_1, p_2, \dots, p_K]$ drawn from $\text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_K)$

- Mean: $\mathbb{E}[p_k] = \frac{\alpha_k}{\sum_{k=1}^K \alpha_k}$

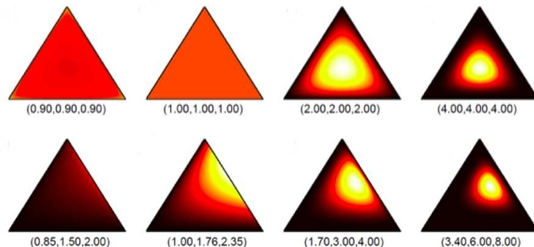
- Variance: $\text{var}[p_k] = \frac{\alpha_k(\alpha_0 - \alpha_k)}{\alpha_0^2(\alpha_0 + 1)}$ where $\alpha_0 = \sum_{k=1}^K \alpha_k$

- Note: $\mathbf{p}$ is a point on $(K - 1)$ -simplex

- Note: $\alpha_0 = \sum_{k=1}^K \alpha_k$ controls how peaked the distribution is

- Note: α_k 's control where the peak(s) occur

Plot of a 3 dim. Dirichlet (2 dim. simplex) for various values of α :

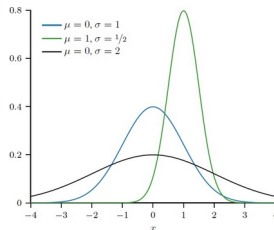


Now comes the
Gaussian (Normal) distribution..

Univariate Gaussian Distribution

- Distribution over real-valued scalar r.v. x
- Defined by a scalar **mean** μ and a scalar **variance** σ^2
- Distribution defined as

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

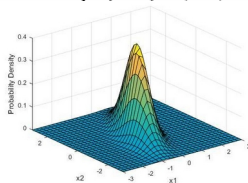


- Mean: $E[x] = \mu$
- Variance: $\text{var}[x] = \sigma^2$
- Precision (inverse variance) $\beta = 1/\sigma^2$

Multivariate Gaussian Distribution

- Distribution over a multivariate r.v. vector $\mathbf{x} \in \mathbb{R}^D$ of real numbers
- Defined by a **mean vector** $\boldsymbol{\mu} \in \mathbb{R}^D$ and a $D \times D$ **covariance matrix**

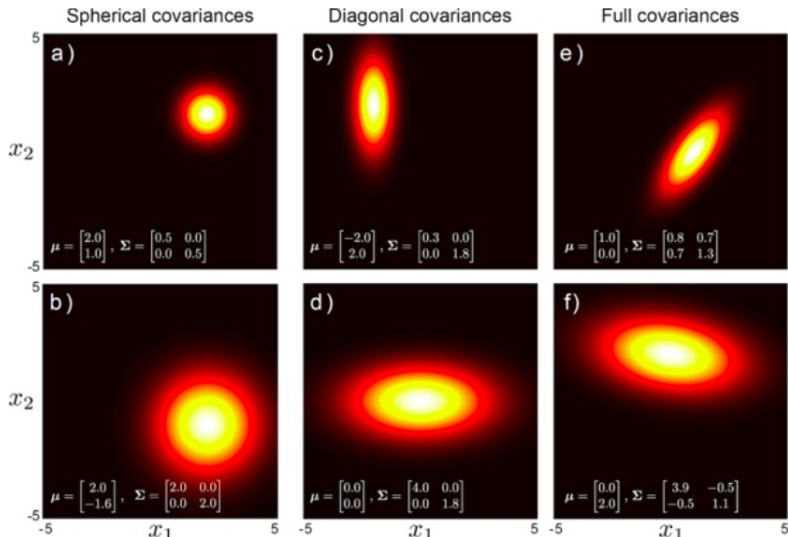
$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$



- The covariance matrix $\boldsymbol{\Sigma}$ must be symmetric and positive definite
 - All eigenvalues are positive
 - $\mathbf{z}^\top \boldsymbol{\Sigma} \mathbf{z} > 0$ for any real vector $\mathbf{z}$
- Often we parameterize a multivariate Gaussian using the inverse of the covariance matrix, i.e., the **precision matrix** $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$

Multivariate Gaussian: The Covariance Matrix

The covariance matrix can be spherical, diagonal, or full



Some nice properties of
the Gaussian
distribution..

Multivariate Gaussian: Marginals and Conditionals

- Given $\mathbf{x}$ having multivariate Gaussian distribution $N(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$. Suppose

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}, \quad \boldsymbol{\Lambda} = \begin{pmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} \\ \boldsymbol{\Lambda}_{ba} & \boldsymbol{\Lambda}_{bb} \end{pmatrix}$$

- The marginal distribution is simply

$$p(\mathbf{x}_a) = N(\mathbf{x}_a | \boldsymbol{\mu}_a, \boldsymbol{\Sigma}_{aa})$$

- The conditional distribution is given by

$$\begin{aligned} p(\mathbf{x}_a | \mathbf{x}_b) &= \mathcal{N}(\mathbf{x}_a | \boldsymbol{\mu}_{a|b}, \boldsymbol{\Lambda}_{aa}^{-1}) \\ \boldsymbol{\mu}_{a|b} &= \boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ab} (\mathbf{x}_b - \boldsymbol{\mu}_b) \end{aligned}$$

**Thus marginals and conditionals
of Gaussians are Gaussians**

Multivariate Gaussian: Marginals and Conditionals

- Given the conditional of an r.v. $\mathbf{y}$ and marginal of r.v. $\mathbf{x}$, $\mathbf{y}$ is conditioned on

$$\begin{aligned}p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{Ax} + \mathbf{b}, \mathbf{L}^{-1}) \\p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})\end{aligned}$$

- Marginal of $\mathbf{y}$ and “reverse” conditional are given by

$$\begin{aligned}p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\Sigma}\{\mathbf{A}^T\mathbf{L}(\mathbf{y} - \mathbf{b}) + \boldsymbol{\Lambda}\boldsymbol{\mu}\}, \boldsymbol{\Sigma}) \\p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^T)\end{aligned}$$

where $\boldsymbol{\Sigma} = (\boldsymbol{\Lambda} + \mathbf{A}^T\mathbf{L}\mathbf{A})^{-1}$

- Note that the “reverse conditional” $p(\mathbf{x}|\mathbf{y})$ is basically the posterior of $\mathbf{x}$ if the prior is
- $p(\mathbf{x})$ Also note that the marginal $p(\mathbf{y})$ is the predictive distribution of $\mathbf{y}$ after integrating
- out $\mathbf{x}$

Very useful property for probabilistic models with Gaussian likelihoods and/or priors.

Gaussians: Product of Gaussians

- Pointwise multiplication of two Gaussians is another (unnormalized) Gaussian

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \mathcal{N}(\mathbf{x}; \boldsymbol{\nu}, \mathbf{P}) = \frac{1}{Z} \mathcal{N}(\mathbf{x}; \boldsymbol{\omega}, \mathbf{T}),$$

where

$$\mathbf{T} = (\boldsymbol{\Sigma}^{-1} + \mathbf{P}^{-1})^{-1}$$

$$\boldsymbol{\omega} = \mathbf{T}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \mathbf{P}^{-1}\boldsymbol{\nu})$$

$$Z^{-1} = \mathcal{N}(\boldsymbol{\mu}; \boldsymbol{\nu}, \boldsymbol{\Sigma} + \mathbf{P}) = \mathcal{N}(\boldsymbol{\nu}; \boldsymbol{\mu}, \boldsymbol{\Sigma} + \mathbf{P})$$

Multivariate Gaussian: Linear Transformations

- Given a $\mathbf{x} \in \mathbb{R}^d$ with a multivariate Gaussian distribution

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

- Consider a linear transform of $\mathbf{x}$ into $\mathbf{y} \in \mathbb{R}^D$

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$$

where $\mathbf{A}$ is $D \times d$ and $\mathbf{b} \in \mathbb{R}^D$

- $\mathbf{y} \in \mathbb{R}^D$ will have a multivariate Gaussian distribution

$$\mathcal{N}(\mathbf{y}; \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$$

Some Other Important Distributions

- **Wishart** Distribution and **Inverse Wishart** (IW) Distribution: Used to model $D \times D$ p.s.d. matrices
 - Wishart often used as a conjugate prior for modeling precision matrices, IW for covariance matrices For $D = 1$, Wishart is the same as gamma dist., IW is the same as inverse gamma (IG) dist.
- **Normal-Wishart** Distribution: Used to model mean and precision matrix of a multivar. Gaussian
- **Normal-Inverse Wishart (NIW)**: : Used to model mean and cov. matrix of a multivar. Gaussian For $D = 1$, the corresponding distr. are **Normal-Gamma** and **Normal-Inverse Gamma (NIG)**

Student-t Distribution (a more robust version of Normal distribution)

Can be thought of as a mixture of infinite many Gaussians with different precisions (or a single Gaussian with its precision/precision matrix given a gamma/Wishart prior and integrated out)